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A Pruning Method via Joint Filter Diversity and Contribution Evaluation

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Abstract—Filter pruning, as an effective structured model compression technique, can significantly reduce the computational complexity and storage overhead of deep convolutional neural networks, thereby facilitating their deployment on resource-constrained edge devices. However, most existing pruning methods evaluate filter importance from a single perspective, such as filter-intrinsic properties (filter weights) or activation-based responses, which makes it difficult to comprehensively assess the true contribution of filters to model performance. To address this limitation, we propose a pruning method based on joint filter diversity and contribution evaluation (JDCE). The method constructs a comprehensive and balanced criterion for filter importance by integrating linear redundancy in the parameter space with task-driven gradient response information. Extensive experiments on CIFAR and ImageNet demonstrate that the proposed method achieves high compression ratios on architectures such as ResNet and VGG while maintaining or even improving classification accuracy, outperforming several state-of-the-art pruning methods.

Index Terms—Model Compression, Deep Neural Networks, Structured Pruning, Filter Pruning

I. INTRODUCTION

The rapid advancement of Convolutional Neural Networks (CNNs) has driven significant progress in computer vision, achieving outstanding performance in image classification [1], [2], object detection [3], [4], and semantic segmentation [5], [6]. Nevertheless, such performance gains come at the cost of rapidly increasing model complexity. As network depth and width expand, CNNs contain an excessive number of parameters and demand substantial computational resources, making deployment on resource-constrained and power-sensitive devices particularly challenging. To address this challenge, researchers have proposed various model compression and acceleration techniques aimed at simplifying models and reducing computational costs. To address this challenge, researchers have proposed various model compression and acceleration

techniques aimed at simplifying the model and reducing the computational cost.

Among various types of model compression methods [7]–[11], network pruning [11] has attracted much attention because of its intuitive and effective. Among them, structural pruning [12]–[15] (e.g., filter or channel pruning) can directly generate hardware-friendly compact architectures by removing redundant structural units, thereby significantly improving inference efficiency. The core of this technique lies in accurately evaluating the importance of each filter to identify and remove redundant filters that have the least impact on model performance.

Existing criteria for evaluating filter importance can be broadly categorized into two groups, filter-intrinsic properties and activation-based responses. The first group relies on filter-intrinsic properties, such as weight norms [16], [17] or the geometric median [12], based on the assumption that filters with smaller weight magnitudes contribute less to the network. The second group focuses on activation-based responses, employing metrics such as feature map rank [14] or Taylor expansion-based criteria [18] to estimate filter importance according to their activation responses. However, these approaches typically adopt a single perspective: filter-intrinsic properties, weight-based methods fail to reflect a filter's sensitivity to actual data distributions, whereas activation-based approaches may overlook intrinsic parametric redundancy.

To address these limitations, we propose a novel filter pruning method based on joint filter diversity and contribution evaluation. The core idea is that effective pruning criteria should capture both a filter's intrinsic redundancy in the parameter space and its sensitivity to the loss. Specifically, we integrate linear representation error with gradient-based information to construct a bidirectional evaluation scheme, as shown in Fig. 1.

Furthermore, pruning inevitably introduces information loss and performance degradation. To rapidly stabilize the model

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post-pruning and establish a strong foundation for subsequent fine-tuning, we design an error-aware weight adjustment strategy. This involves inserting a convolution layer, initialized as an identity mapping, after a pruned layer. Its weights are then dynamically adjusted based on the linear representation coefficients, thereby smoothly redistributing the functionality of pruned filters to the surviving ones and effectively controlling the resultant feature map error.

To summarize, our main contributions are three-fold:

- 1) Propose a pruning method jointing filter diversity and contribution evaluation, which solves the 'single view bias' problem of existing methods.
- 2) Propose a bidirectional filter evaluation criterion. This criterion integrates parameter-space redundancy and gradient-based contribution for more precise importance scoring.
- 3) Design an error-aware weight adjustment strategy. This strategy dynamically configures convolutional weights to mitigate feature map error induced by pruning, thereby reducing information loss.

II. RELATED WORK

Depending on whether the evaluation process relies on input data to observe the "behavior" of filters, existing research of network pruning can be broadly categorized into two classes: methods based on filter-intrinsic properties and activation-based responses.

A. Methods Based on Filter-intrinsic Properties

This category of methods relies solely on the pretrained model's weight parameters themselves, judging filter importance by analyzing inherent static characteristics. Their advantage is that they require no input data, resulting in high computational efficiency.

Norm-based Methods. This is one of the most intuitive and widely adopted criteria. The fundamental assumption is that filters with smaller weight norms extract less important features. For instance, the pioneering work by Li et al. [16] proposed ranking filters using the L1 norm and pruning those with the smallest norms. While such methods are computationally simple, critics argue that the relationship between weight magnitude and filter importance is not necessarily linear, which may lead to the erroneous removal of some filters that, despite having small weights, possess discriminative power.

Structural Redundancy-based Methods. These methods evaluate importance from the perspective of similarity or replaceability among filters. He et al. [12] proposed pruning based on the geometric median (PFGM), positing that a filter is redundant if it can be well-represented by the geometric median of other filters in its layer. The Linear Replaceable Filter (LRF) method proposed by Joo et al. [19] goes a step further. It precisely quantifies a filter's redundancy by calculating the extent to which it can be linearly represented by other filters in the same layer (i.e., the linear representation error). Our work draws inspiration from such methods, but the innovation lies in not treating this static redundancy as the sole

criterion. Instead, we incorporate it as a key dimension within our bidirectional evaluation framework.

B. Methods Based on Activation-Based Responses

This category of methods requires a subset of input data. They assess importance by observing a filter's activation-based responses—such as activation values or gradients—during forward or backward propagation. This approach better reflects a filter's actual contribution to a specific task and data distribution.

Activation-Values Methods. These methods make judgments based on output feature maps (activations). For example, the HRank method proposed by Lin et al. statistically determines, using a large number of input images, that filters corresponding to low-rank feature maps contain less information and can thus be pruned. Xie et al. [20], from a model interpretability perspective, leverage class activation maps combined with discrete wavelet transform to assess the spatial reliability of channels, which also relies on dynamic information from forward propagation.

Sensitivity-based Methods. These methods evaluate importance by analyzing the sensitivity of the loss function to changes in filter weights, typically embodied in gradient form. Molchanov et al. [18] utilized first-order Taylor expansion to approximate the loss change caused by removing each filter. The work in [20] also employed gradient information to enhance feature maps for a better understanding of channel contribution. Our method inherits this idea, treating gradient information as a direct representation of a filter's dynamic contribution and integrating it into our evaluation framework.

Our method aims to fuse the advantages of both approaches. We simultaneously utilize gradient information (a core metric of contribution) and linear representation error (a precise quantification of redundancy). Through normalization and weighted fusion, we achieve a more comprehensive and robust assessment of filter importance.

III. METHOD

A. Problem Definition

Given a pre-trained convolutional neural network with N convolutional layers, denoted as $\mathcal{L} = \{L_1, L_2, \dots, L_N\}$. For the n -th layer, the convolutional kernel parameters $\mathcal{L}_n = \{f_n^1, f_n^2, \dots, f_n^{c_n}\}$ form a four-dimensional tensor of size $c_n \times c_{n-1} \times k \times k$, where c_n is the number of filters in the n -th layer, c_{n-1} is the number of input channels (i.e., the number of filters in the $(n-1)$ -th layer), and $k \times k$ is the kernel size. $f_n^{i,j}$ represents the j -th channel of the i -th filter in the n -th layer. The goal of this paper is to identify and remove weakly expressive or redundant filters, thereby constructing an optimal feature subset to reduce the model's storage requirements and computational complexity.

B. Filter bidirectional evaluation criteria

Our framework supports pruning at both the filter and channel-level granularities. For simplicity, we describe the

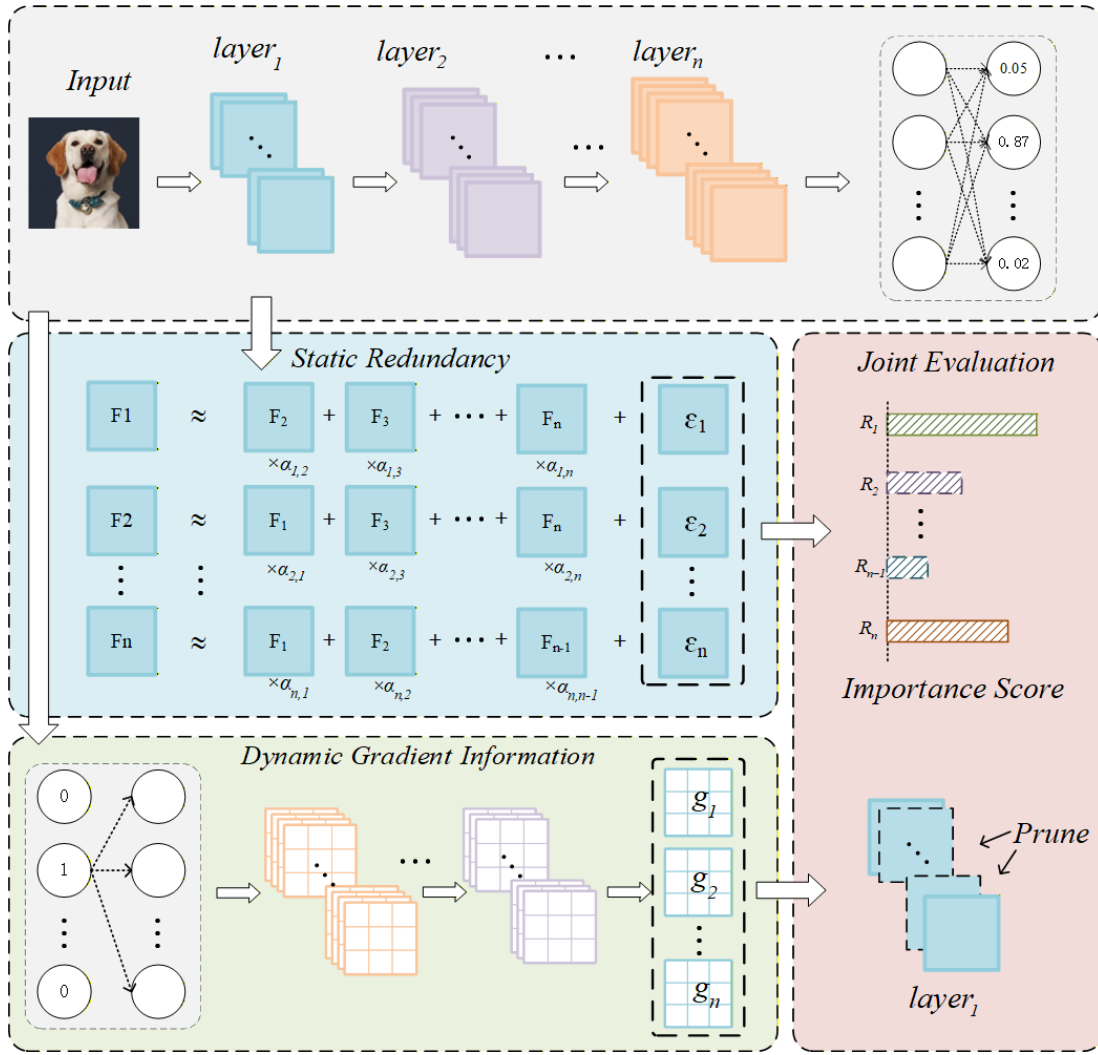


Fig. 1: Framework of JDCE. The framework consists of four main components: 1)Input layers: Representing multiple network layers ($layer_1$ to $layer_n$); 2)Filter-intrinsic properties: Characterizing redundancy relationships among filters within the same layer through a linear approximation model, where F_i can be approximated by weighted sums of other filters plus residual ε_i , with coefficients $\alpha_{i,j}$ reflecting inter-layer dependencies; 3)Activation-based responses: Incorporating gradient signals $g_1, g_2, \dots, g_n$ from training to enhance the temporal adaptability of importance evaluation, with particular attention to gradient propagation in $layer_1$. 4)Joint evaluation module: Generating redundancy scores $R_1, R_2, \dots, R_n$ and aggregating them into final importance scores;More details can be found in Section III

filter-level pruning here. The pruning objective can be formulated as the following optimization problem:

$$\max_{\delta_{n,i}} \sum_{j=1}^N \sum_{i=1}^{c_n} \delta_{n,i} R(f_n^i) \quad \text{s.t.} \quad \sum_{i=1}^{c_n} \delta_{n,i} = (1-r) \cdot c_n, \quad (1)$$

where $\delta_{n,i} \in \{0, 1\}$ is the retention indicator for the i -th filter in the n -th layer (1 for keep, 0 for prune), r is the preset pruning rate, and the constraint ensures that $(1-r)c_n$ filters are kept in this layer. $R(\cdot)$ is the filter importance metric function, whose design is one of the core contributions of this paper.

Inspired by recent advances in pruning, we recognize that jointly considering the input data and the filter helps to find a

better network structure. Therefore, we introduce filter gradient as a importance measure. Gradients intuitively reflect the sensitivity of filters to input data. By dynamically computing gradients using a small batch of training data, our method better adapts to the actual data distribution, providing finer pruning and reducing the risk of overfitting compared to strategies relying on global statistics.

Let the input to the n -th layer be $I_n = \{I_n^1, I_n^2, \dots, I_n^{c_{n-1}}\} \in \mathbb{R}^{b \times c_{n-1} \times h_n \times w_n}$, where b is the batch size, h_n and w_n are the height and width of the feature maps, respectively. The convolutional operation of this layer can be expressed as follows.

$$X_n = L_n(I_n) = \sum_{i=1}^{c_n-1} I_n^i * f_n^i, \quad (2)$$

where X_n is the output feature map and $*$ denotes the 2D convolution operation.

Through forward and backward propagation, we obtain the gradients of each convolution layer: $\text{grad} = \{\text{grad}_1, \text{grad}_2, \dots, \text{grad}_N\}$, where $\text{grad}_n \in \mathbb{R}^{c_n \times c_{n-1} \times k \times k}$. The magnitude of the absolute gradient reflects the sensitivity of the filter weights to the loss function. A larger gradient indicates that the weight has a more significant impact on the model output; a smaller gradient suggests lower activity, meaning that even large changes have little effect on the loss. Therefore, we use the sum of absolute gradients of filter weights as a measure of its dynamic contribution:

$$g_n^k = \sum_{i=1}^{c_n-1} \sum_{j=1}^H |\text{grad}_i^j|, \quad (3)$$

where g_n^k denotes the gradient contribution of the k -th filter in the n -th layer, and H is the number of elements in the gradient tensor.

Simultaneously, to evaluate the filter-intrinsic properties, we introduce the linear representation error. For filter f_n^k , we attempt to approximate it by a linear combination of the remaining $c_n - 1$ filters in the same layer: $\{f_n^1, \dots, f_n^{k-1}, f_n^{k+1}, \dots, f_n^{c_n}\}$:

$$f_n^k = \sum_{i \neq k} a_{k,i} f_n^i + \varepsilon_k = F \cdot a_k + \varepsilon_k, \quad (4)$$

where F is the filter matrix with f_n^k removed, $a_k = [a_{k,1}, \dots, a_{k,k-1}, a_{k,k+1}, \dots, a_{k,c_n}]^T$ is the coefficient vector, and ε_k is the approximation error. To minimize the error, we solve the following optimization problem:

$$\min \|\varepsilon_k\|^2 = \min \|f_n^k - F \cdot a_k\|^2. \quad (5)$$

This is solved via the normal equation or pseudo-inverse:

$$a_k = \begin{cases} (F^T F)^{-1} F^T f_n^k, & \text{if } F^T F \text{ is invertible,} \\ F^+ f_n^k, & \text{otherwise,} \end{cases} \quad (6)$$

where F^+ denotes the Moore-Penrose pseudo-inverse of F . After obtaining a_k , we compute the error ε_k . A smaller ε_k indicates that the filter is more easily linearly replaceable by others, i.e., higher redundancy.

To combine the dynamic contribution g_n^k and the static redundancy $\|\varepsilon_k\|$, we first normalize both:

$$\tilde{g}_n^k = \frac{g_n^k - g_{\min}}{g_{\max} - g_{\min}}, \quad \tilde{\varepsilon}_n^k = \frac{\|\varepsilon_k\| - \varepsilon_{\min}}{\varepsilon_{\max} - \varepsilon_{\min}}. \quad (7)$$

After normalization, both values lie in $[0, 1]$. Then, we define the filter importance metric $R(\cdot)$ as:

$$R(f_n^k) = \frac{2 \cdot \tilde{g}_n^k \cdot \tilde{\varepsilon}_n^k}{\tilde{g}_n^k + \tilde{\varepsilon}_n^k + \gamma}, \quad (8)$$

where γ is a small constant (e.g., 10^{-8}) to prevent division by zero. We adopt the harmonic mean formulation to fuse diversity scores and contribution scores, which possesses an inherent property of being highly sensitive to small values, effectively enforcing a strict 'AND' condition. In our pruning context, an ideal filter is required to satisfy simultaneous criteria: it must possess high task relevance (high gradient contribution) and distinct feature representation (low redundancy).

Finally, we prune the filter with the lowest score based on the importance scores computed in Eq. (9):

$$s = \arg \min_i R(f_n^i). \quad (9)$$

C. Error-aware Weight Adjustment Strategy

Pruning reduces parameters but inevitably loses some feature information. To mitigate this, we propose an error-aware weight adjustment strategy based on linear representation.

From Eq. (4) and the linearity of convolution, the output feature map X_n^k corresponding to the pruned filter f_n^k can be approximated as:

$$X_n^k \approx \sum_{j \neq k} a_{k,j} X_n^j, \quad (10)$$

where X_n^j are the feature maps produced by the remaining filters. This implies that the information loss due to pruning is $\Delta X \approx X_n^k$.

To compensate for this loss, we insert a 1×1 convolutional layer $W_{1 \times 1}$ after the target convolutional layer and initialize it as an identity matrix, ensuring that the insertion does not alter the initial network output. After pruning filter f_n^k , we adjust the weights of this 1×1 convolutional layer according to the linear representation coefficients $a_{k,j}$:

$$w'_j = w_j + a_{k,j} \cdot w_k, \quad (11)$$

where w_j and w_k are the 1×1 convolutional weights for the corresponding channels. This operation effectively redistributes the function of the pruned filter to the remaining filters in proportion to the linear coefficients, thereby compensating for the error at the feature map level.

D. Overall Pruning Process

The complete pruning pipeline proceeds in four consecutive stages. 1) The pre-trained model is loaded and a small batch of training data is prepared for initialization. 2) Forward and backward propagation are performed to compute the gradient contribution for each filter, while the linear representation problem is solved to obtain its redundancy error; these two metrics are then normalized and fused into a unified importance score. 3) An iterative pruning and compensation stage is executed, wherein the filter with the lowest importance score is sequentially removed, followed by an error-aware weight adjustment that updates the accompanying 1×1 convolutional

TABLE I: Pruning results on the CIFAR-10 dataset. Δ Top-1(the bigger, the better) is the accuracy difference of pruned model compared to the baseline model. FLOPs↓ and Param↓ represent the reduction of FLOPs and parameters in percentage, respectively.

Model	Algorithm	Baseline(%)	Top-1(%)	Δ Top-1(%)	FLOPs ↓	Params ↓
ResNet-32	DCPH [21]	93.34	92.85	- 0.49 ↓	30.00%	N/A
	FPGM [12]	92.63	92.31	+ 0.32 ↑	41.50%	N/A
	SFP [17]	92.63	92.08	- 0.55 ↓	41.50%	N/A
	TAS [22]	93.89	93.16	- 0.73 ↓	49.40%	N/A
	JDCE(Ours)	92.46	92.97	+ 0.51 ↑	62.00%	63.30%
	JDCE(Ours)	92.46	92.27	- 0.19 ↓	73.50%	74.00%
ResNet-56	HRank [14]	93.26	93.85	+ 0.59 ↑	28.00%	22.30%
	CFDP [23]	93.26	93.97	+ 0.71 ↑	28.00%	22.30%
	DCFF [24]	93.26	93.26	0	55.90%	56.00%
	ResPrune [25]	93.59	93.51	- 0.08 ↓	57.98%	N/A
	TAILOR [26]	93.26	93.38	+ 0.12 ↑	61.22%	55.29%
	JDCE(Ours)	93.45	94.12	+ 0.67 ↑	62.40%	63.43%
	FPICI [27]	93.26	92.49	- 0.77 ↓	73.30%	74.10%
	ASTER [28]	93.80	92.79	- 1.01 ↓	73.40%	N/A
	HSC [29]	93.26	92.80	- 0.46 ↓	73.00%	74.00%
	JDCE(Ours)	93.45	93.60	+ 0.15 ↑	73.90%	74.12%
ResNet-110	APIB [30]	93.26	93.92	+ 0.66 ↑	54.00%	50.00%
	HRank [14]	93.50	93.36	- 0.14 ↓	58.20%	59.20%
	JDCE(Ours)	93.76	94.50	+ 0.74 ↑	62.63%	63.51%
	ResPrune [25]	93.68	94.19	+ 0.51 ↑	63.43%	N/A
	DCFF [24]	93.50	93.80	+ 0.30 ↑	66.60%	N/A
	TAILOR [26]	93.50	93.81	+ 0.31 ↑	67.18%	68.02%
	HSC [29]	93.50	93.56	+ 0.06 ↑	71.80%	70.00%
	ASTER [28]	93.93	94.07	+ 0.14 ↑	72.10%	N/A
	JDCE(Ours)	93.76	94.17	+ 0.41 ↑	74.10%	74.20%
VGG-16	ResPrune [25]	93.76	94.03	+ 0.27 ↑	57.28%	N/A
	FPICI [27]	93.96	94.30	+ 0.34 ↑	58.90%	83.1%
	JDCE(Ours)	93.02	93.66	+ 0.64 ↑	63.00%	64.50%
	ASTER [28]	93.90	94.21	+ 0.31 ↑	63.70%	N/A
	FalCon [31]	93.32	91.92	- 1.40 ↓	67.30%	N/A
	HSC [29]	93.96	93.75	- 0.21 ↓	69.30%	86.00%
	JDCE(Ours)	93.02	93.21	+ 0.19 ↑	74.00%	74.74%

layer weights to compensate for the removal; this process repeats until a predefined layer-wise pruning rate is met. 4) A global fine-tuning stage is applied to the fully pruned model to recover and further enhance its performance.

IV. EXPERIMENTS

A. Experimental Setup

To evaluate the effectiveness of the proposed filter pruning method, we conduct extensive experiments on two widely used benchmark datasets: CIFAR-10 [32] and ImageNet [33]. CIFAR-10 contains 50,000 training images and 10,000 test images from 10 categories, with a resolution of 32×32 . ImageNet is a large-scale image classification benchmark consisting of over 1.2 million training images and 50,000 validation images across 1,000 classes, where images are resized to 224×224 .

We evaluate our method on both single-branch and multi-branch network architectures, including VGG-16 [34], ResNet-32, ResNet-56, ResNet-110 [35] on CIFAR-10, as well as ResNet-18 and ResNet-50 on ImageNet. All models are implemented in PyTorch. For CIFAR-10, standard data augmentation strategies such as random cropping and horizontal flipping are applied. For ImageNet, common preprocessing techniques

including resizing, random cropping and normalization are used.

During pruning, gradient information is computed using a randomly selected mini-batch of 128 images. After pruning, the resulting models are fine-tuned using stochastic gradient descent (SGD). Unless otherwise stated, for CIFAR-10, we fine-tune the pruned models for 300 epochs with a batch size of 128, momentum of 0.9 and weight decay of 0.005. For ImageNet, fine-tuning is performed for 200 epochs with a batch size of 256, momentum of 0.9 and weight decay of $1e-4$. All experiments are conducted on an NVIDIA A800 GPU.

We report Top-1 accuracy for CIFAR-10 and both Top-1 and Top-5 accuracy for ImageNet, together with the reduction ratios of FLOPs and parameters. Note that the parameters and computational costs of the introduced compensation layer are strictly included in the reported statistics. All reported results are averaged over three independent runs.

B. Results on CIFAR-10

We first compare the proposed method with several state-of-the-art filter pruning approaches on CIFAR-10. The results are summarized in Table I.

TABLE II: Pruning results on the ImageNet dataset. Δ Top-1 and Δ Top-5 denote the accuracy difference of the pruned model compared to the baseline model.

Model	Algorithm	Top-1(%)	Δ Top-1(%)	Top-5(%)	Δ Top-5(%)	FLOPs ↓	Params ↓
ResNet-18	MIL [36]	69.98 → 66.33	- 3.65 ↓	89.24 → 86.94	- 2.30 ↓	34.6%	N/A
	SFP [17]	70.28 → 67.10	- 3.18 ↓	89.63 → 87.78	- 1.85 ↓	41.8%	N/A
	FPGM [12]	70.28 → 67.78	- 2.50 ↓	89.63 → 88.01	- 1.62 ↓	41.8%	N/A
	LRMF [37]	70.28 → 67.87	- 2.41 ↓	89.63 → 87.87	- 1.76 ↓	53.3%	N/A
	JDCE(Ours)	69.72 → 69.43	- 0.29 ↓	89.09 → 89.21	+ 0.12 ↑	56.9%	59.7%
ResNet-50	MFP [38]	76.15 → 75.67	- 0.48 ↓	92.87 → 92.81	- 0.06 ↓	42.2%	N/A
	HRank [14]	76.15 → 74.98	- 1.17 ↓	92.87 → 92.33	- 0.54 ↓	43.7%	36.6%
	CPS [39]	76.15 → 75.59	- 0.56 ↓	N/A	N/A	44.3%	N/A
	DCFF [24]	76.15 → 75.18	- 0.97 ↓	92.96 → 92.56	- 0.40 ↓	45.3%	40.7%
	FalCon [31]	75.83 → 74.59	- 1.24 ↓	92.78 → 92.51	- 0.27 ↓	53.5%	N/A
	ABCPruner [40]	76.01 → 73.86	- 2.15 ↓	92.96 → 91.69	- 1.27 ↓	54.3%	N/A
	HSC [29]	76.01 → 75.52	- 0.63 ↓	92.87 → 92.65	- 0.22 ↓	57.3%	N/A
	APIB [30]	76.15 → 76.07	- 1.24 ↓	N/A	N/A	56.0%	50.0%
	SASL [41]	76.15 → 75.15	- 1.00 ↓	92.87 → 92.58	- 0.40 ↓	56.1%	50.0%
	ResPrune [25]	76.15 → 75.10	- 1.05 ↓	N/A	N/A	58.9%	N/A
	AutoPruner [42]	76.15 → 74.76	- 1.39 ↓	92.87 → 92.15	- 0.72 ↓	51.2%	N/A
	JDCE(Ours)	76.00 → 75.88	- 0.12 ↓	92.92 → 92.85	- 0.07 ↓	51.6%	53.7%

For ResNet-32, our method consistently outperforms existing pruning methods under different compression ratios. At a pruning ratio of 0.5, our method improves the Top-1 accuracy from 92.46% to 92.97%, while reducing FLOPs and parameters by 62.0% and 63.3%, respectively. In contrast, competing methods such as FPGM [12], TAS [22] and DCPH [21] lead to noticeable accuracy degradation under similar compression levels. Even at a higher pruning ratio of 0.6, our method maintains competitive accuracy with only a 0.19% drop while achieving over 73% reduction in both FLOPs and parameters.

Similar trends are observed on deeper networks. On ResNet-56 and ResNet-110, the proposed method not only achieves substantial reductions in computational cost but also leads to accuracy improvements over the unpruned baseline at moderate pruning ratios. This demonstrates that redundancies in deeper networks can be effectively removed by jointly considering gradient sensitivity and redundancy estimation.

For the single-branch VGG-16 network, our method also shows clear advantages. Compared with existing pruning approaches, the proposed method achieves higher accuracy under the same compression ratio, indicating that it generalizes well across different network structures.

Overall, the results on CIFAR-10 demonstrate that the proposed method achieves a favorable trade-off between model accuracy and computational efficiency across various architectures and pruning ratios.

C. Results on ImageNet

To further verify the scalability of the proposed method, we conduct experiments on the large-scale ImageNet dataset. The quantitative results are reported in Table II.

ResNet-18. Our method performs excellently in preserving model performance. Compared to the baseline, it only incurs a 0.29% drop in Top-1 accuracy (from 69.72% to 69.43%), while

the Top-5 accuracy even slightly improves (from 89.09% to 89.21%). In contrast, other methods such as FPGM [12], SFP [17] and LRMF [37] lead to larger performance degradation (Top-1 accuracy drops ranging from 2.41% to 3.18%) under the same conditions. Meanwhile, our method achieves significant reductions in FLOPs and parameters, decreasing them by 56.90% and 59.67%, respectively.

ResNet-50. Our method also demonstrates outstanding performance. Compared to the baseline, it only results in a 0.12% drop in Top-1 accuracy (76.00% → 75.88%), while the Top-5 accuracy remains nearly unchanged (92.92% → 92.85%). In comparison, other methods such as DCFF [24], HSC [29] and APIB [30] lead to more significant performance degradation (Top-1 accuracy drops ranging from 0.63% to 1.24%) under the same conditions. Furthermore, our method achieves notable reductions in FLOPs and parameters, decreasing them by 51.60% and 53.70%, respectively.

In summary, our method exhibits excellent performance on the ImageNet dataset. On both ResNet-18 and ResNet-50, it maintains high model performance while substantially reducing computational and storage costs, further demonstrating the effectiveness and superiority of the proposed method on large-scale datasets.

D. Ablation Study

To verify the effectiveness of each component in JDCE, specifically the joint evaluation metric and the Error-aware Weight Adjustment Strategy (EWAS), we conduct ablation experiments on ResNet-56 and VGG-16 using the CIFAR-10 dataset. The results are reported in Table III.

1) *Impact of Joint Evaluation Metric:* We first evaluate the superiority of the proposed joint metric by comparing it with single-perspective criteria. To ensure a fair comparison, all variants in this group (Var A, Var B, and JDCE) are equipped with the EWAS strategy.

TABLE III: Ablation study of individual components on ResNet-56 with 62.40% FLOPs dropped and VGG-16 with 63.00% FLOPs dropped. $\mathcal{D}$ and $\mathcal{C}$ denote Filter Diversity and Contribution. EWAS is the Error-aware Weight Adjustment Strategy.

Model	Method(Variant)	Metric	EWAS	Top-1 %
ResNet-56	Baseline	-	-	93.45
	Var A	$\mathcal{D}$	✓	93.70
	Var B	$\mathcal{C}$	✓	93.32
	Var C	$\mathcal{D} + \mathcal{C}$	×	93.83
	JDCE (Ours)	$\mathcal{D} + \mathcal{C}$	✓	94.12
VGG-16	Baseline	-	-	93.02
	Var A	$\mathcal{D}$	✓	93.48
	Var B	$\mathcal{C}$	✓	93.16
	Var C	$\mathcal{D} + \mathcal{C}$	×	93.21
	JDCE (Ours)	$\mathcal{D} + \mathcal{C}$	✓	93.66

Single Metric vs. Joint Metric. As shown in Table III, for ResNet-56, pruning based solely on diversity (Var A) or contribution (Var B) yields Top-1 accuracies of 93.70% and 93.32%, respectively. While these single metrics perform reasonably well, they are outperformed by our joint evaluation method (JDCE), which achieves 94.12%.

Consistency. A similar trend is observed on VGG-16, where the joint metric (JDCE) achieves 93.66%, surpassing Var A (93.48%) and Var B (93.16%). This confirms that fusing parameter-space redundancy with task-driven gradient information provides a more robust criterion, effectively identifying filters that are both redundant and less informative.

2) *Impact of Error-aware Weight Adjustment (EWAS)*: We further investigate the necessity of the proposed compensation mechanism by comparing the method with and without EWAS.

ResNet-56. Comparing Var C (Joint metric without EWAS) with the full JDCE method, the introduction of EWAS improves accuracy from 93.83% to 94.12% (+0.29%).

VGG-16. The effect is even more pronounced on VGG-16. Without EWAS (Var C), the accuracy drops to 92.99%, which is slightly below the baseline (93.02%). However, by applying EWAS, the performance is significantly restored and improved to 93.66% (+0.45%).

These results demonstrate that the linear reconstruction-based compensation (EWAS) plays a critical role in mitigating the feature distribution shift caused by pruning, especially for architectures sensitive to structure modification like VGG-16.

V. CONCLUSIONS

In this paper, we present a joint filter diversity and contribution evaluation based filter pruning method for convolutional neural networks (JDCE). By combining gradient sensitivity with linear redundancy estimation, the proposed method can more accurately identify less important or redundant filters for removal. In addition, an information recovery strategy based on 1×1 convolution is introduced to compensate for the feature information lost during pruning, helping the pruned network retain its original functionality. Extensive experiments

on CIFAR-10 and ImageNet across multiple network architectures demonstrate that the proposed method can significantly reduce the number of FLOPs and parameters while maintaining or even improving the classification accuracy, which demonstrates the effectiveness and universality of the method. For example, pruning ResNet-56 on CIFAR-10 can reduce 62.40% FLOPs and 63.43% parameters, and improve the Top-1 accuracy by 0.67%. Pruning ResNet-50 on ImageNet can reduce 51.6% FLOPs and 53.7% parameters, while Top-1 accuracy only decreases 0.12%.

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